

Generating a sine wave

1. Using a single resistor

In this experiment, the objective is to generate a sine wave. There are many waves to generate a sine wave, and 2 most used techniques are the Wein bridge and the timer 555 circuit each used based on the range of frequencies and applications. In this experiment we will using a timer 555 chip and to observe the characteristics of the resulting wave.

Components used:

1. Resistors
2. Inductors
3. Capacitors
4. NE555 (bipolar Timer 555 chip)
5. Voltage supply

The NE555 is a bipolar chip where the range of frequencies best used between 10Hz – 500kHz. In this experiment, I will be using a 1kHz frequency wave. The timer555 outputs a square wave but different components can be used to obtain a sine wave such as RC combination and LC tank. We will be using a LC tank in this experiment in this experiment to obtain a sine wave.

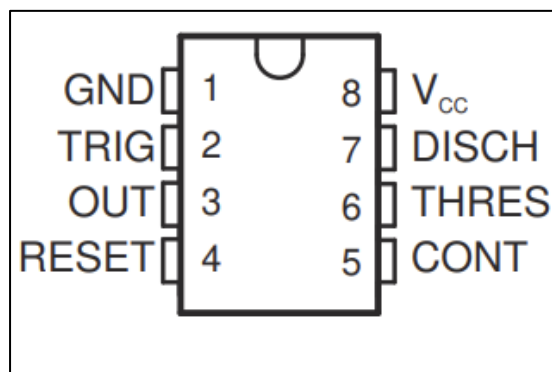


Figure 1: NE555 structure

CONT	5
DISCH	7
GND	1
NC	—
OUT	3
RESET	4
THRES	6
TRIG	2
V _{CC}	8

Figure 2: Pin configuration of NE555

Data extracted from: <https://www.ti.com/lit/ds/symlink/ne555.pdf>

Since 1kHz frequency is used, the values for the resistors and capacitors should be chosen very carefully due to filtering issues and smooth waves. Since the frequency is in kHz range the ideal value of capacitors for this experiment would be in nano-farads range. In this case, the frequency and the capacitance has an inversely proportional relationship. To make sure the capacitor charges and discharges at a stable range capacitors are used in the above range. Accordingly, the other values should be calculated using the below equations.

$$1. \quad f = \frac{1}{2\pi\sqrt{LC}}$$

$$2. \quad f = \frac{1}{2\pi * RC}$$

The choice of the values for the components should be chosen very wisely considering the range of frequency.

RC combination component values:

As mentioned above for the kHz frequency nano range is used for capacitance therefore 10nF was used.

$$1000Hz = \frac{1}{2\pi \times R \times 10nF}$$

$$\frac{1}{1000 \times 2\pi \times 10nF} = R$$

$$R = 15915\Omega \approx 16k\Omega$$

LC tank combination component values:

Considering the frequency, an inductor value in milli range is ideal to be used. 11.5mH was chosen upon many combinations considering availability, ease of finding components.

$$1000Hz = \frac{1}{2\pi\sqrt{11.5 \times 10^{-3} \times C}}$$

$$(1000Hz \times 2\pi)^2 \times 11.5 \times 10^{-3} = \frac{1}{C}$$

$$C = 2.2\mu F$$

Circuit design and simulation

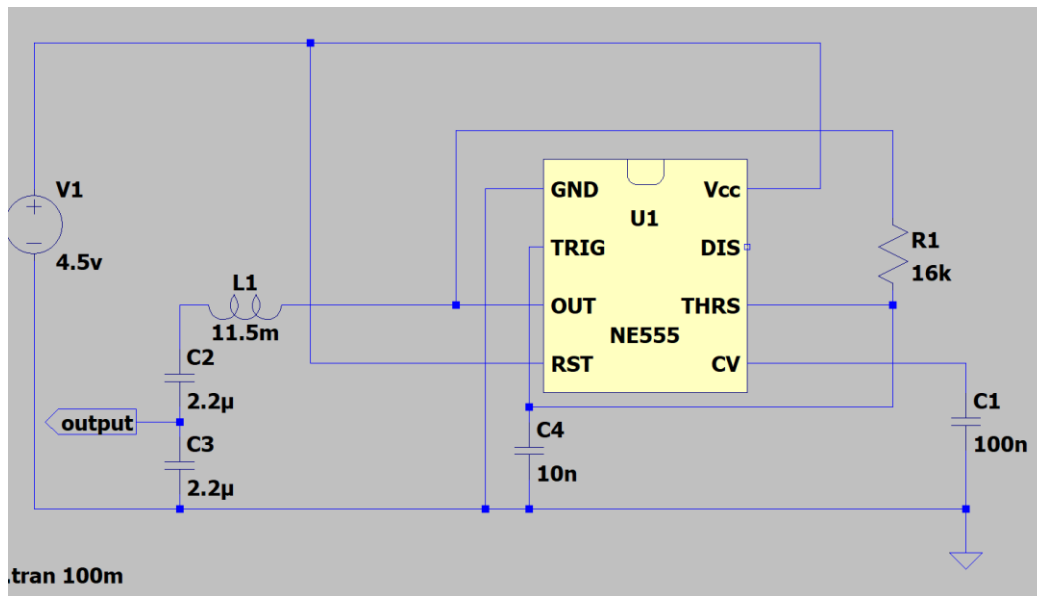


Figure 3: Circuit structure

(Above circuit structure is adapted from: <https://www.learningaboutelectronics.com/Articles/Sine-wave-generator-circuit-with-a-555-timer.php> with modifications)

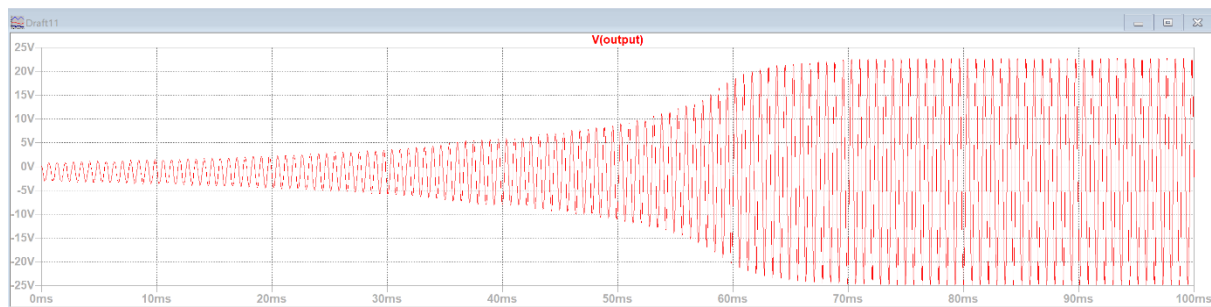


Figure 4: LT spice simulation(0-100ms)



Figure 5: LT spice simulation results(70ms – 92ms zoomed)

Observations in simulations in figure 4 shows that the amplitude gradually increases which is due to the energy built up from oscillations as it starts from 0 and then reaches stability in 65ms.

Importance on choice of components

The circuit below demonstrates regardless of the frequency matches, the choice of components acts an important role in this circuit setup.

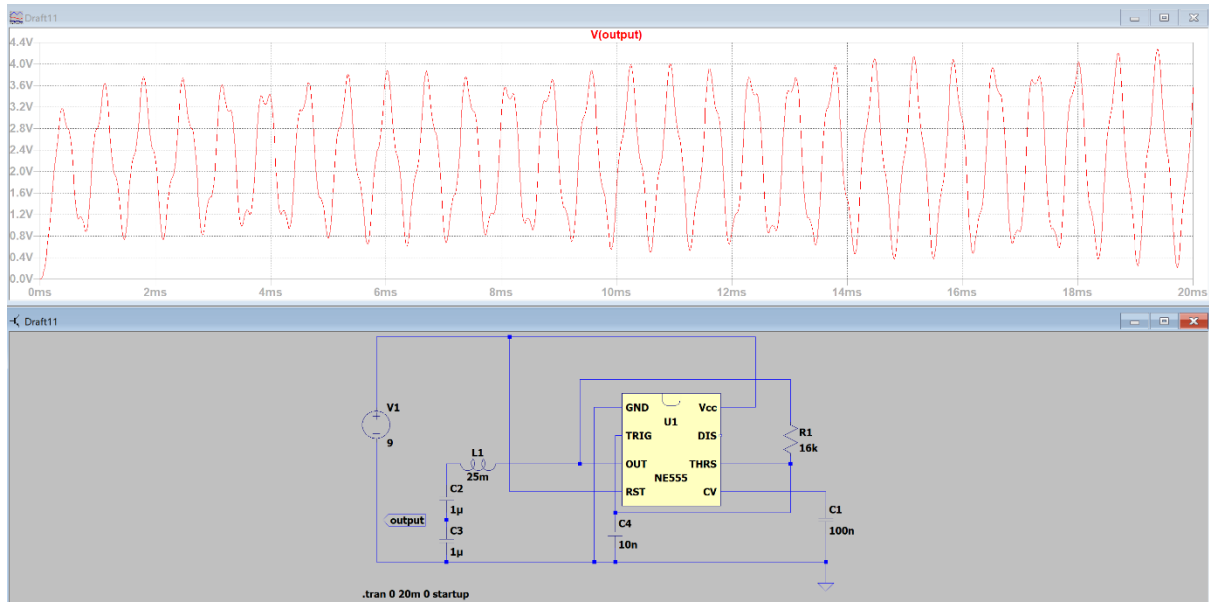


Figure 6: LT spice simulation with component error.

Although the frequencies match approximately to 1 kHz in both the RC and LC tank the capacitors do not filter properly which causes the crests and troughs in the wave not to be smooth hence choice of values plays a very important role in this circuit setup.

In the above setup, the calculated frequency does not match the simulated frequency. This is due to the functionality of the timer 555 has not been used properly.

2. Generating a sine wave using 2 resistors

The timer555 is designed to have 2 resistors to set up proper charging and discharging path for the capacitors. This matches the functionality of the timer 555 hence creating the expected and calculated frequency. The circuit setup and the results are as given below.

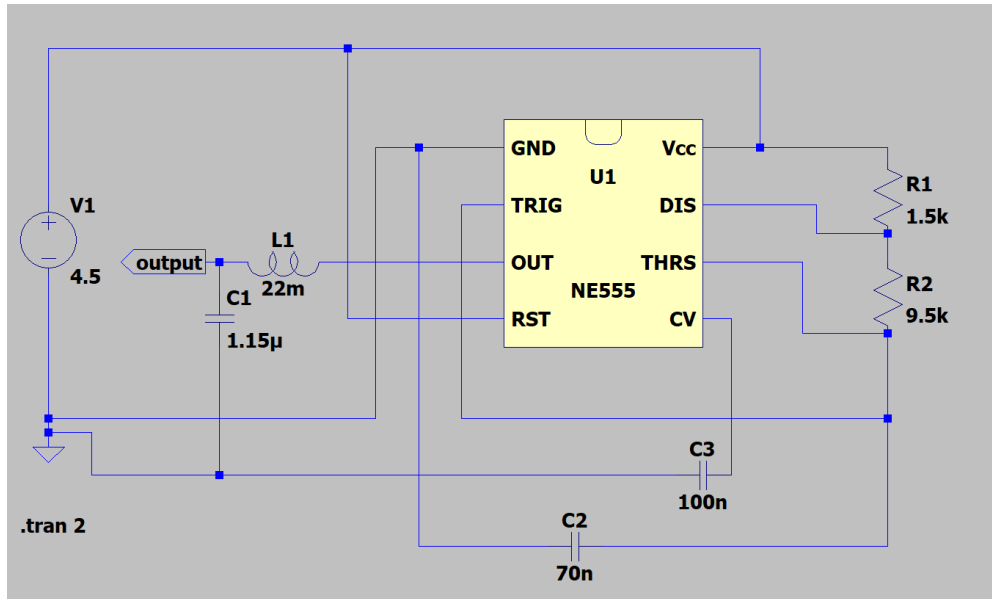


Figure 7: LTspice setup with 2 resistors

Choice of components are considered by the below formulae.

$$f = \frac{1.44}{(R_1 + 2R_2) \times C}$$

$$f = \frac{1}{2\pi\sqrt{LC}}$$

$$D_c = \frac{R_1 + R_2}{R_1 + 2R_2} \times 100$$

Since the same timer 555 chip is being used, the chosen values for the components lies in the same range as in part 1.

The result of the simulation is as below:

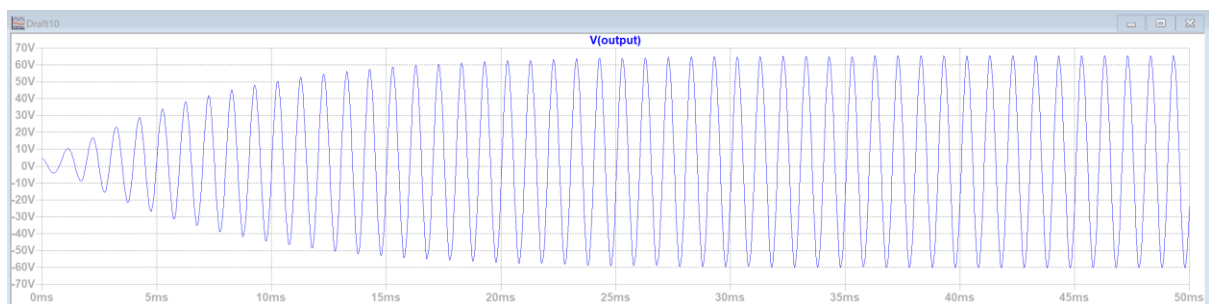


Figure 8: sine wave from 0-50ms

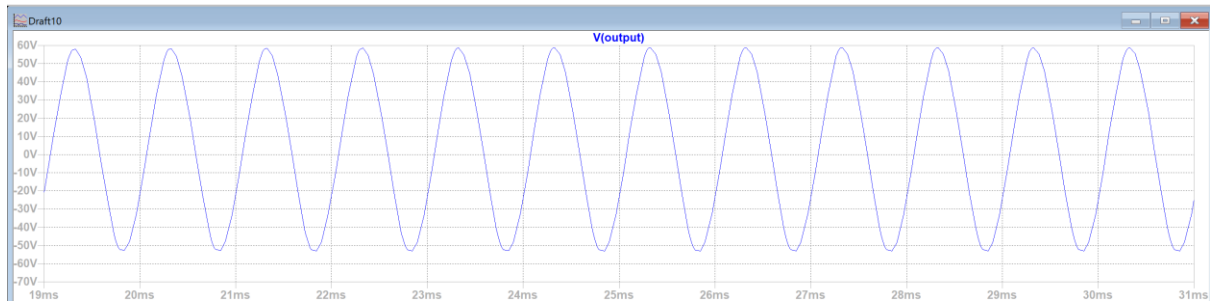


Figure 9: Wave from fig. 8 zoomed to 19ms-31ms

As we can see, the frequency calculated and from the simulation is $1000\text{Hz} \pm 0.5\%$ which is very accurate and precise. In this circuit setup the duty cycle can also be adjusted by the choice of resistors using the above formulae.

A FFT was carried out on LT spice to find if there are any hidden harmonics/to check the purity of the sine wave. The resulted FFT is as below.

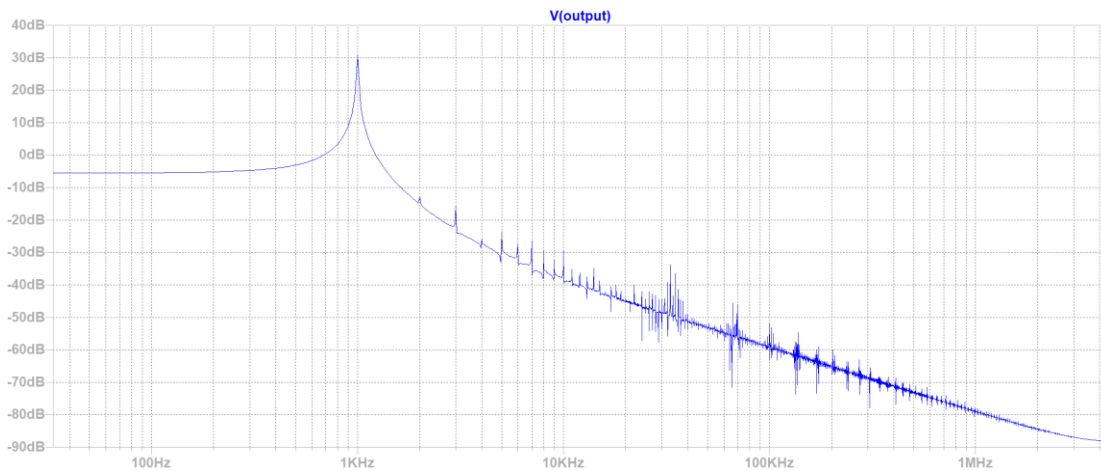


Figure 10: FFT of Figure 9

The FFT was also conducted in MATLAB by extracting the wave data from LTspice and plotting it accordingly onto graph on MATLAB and performing a FFT using the corresponding graph to verify the result in LTspice.

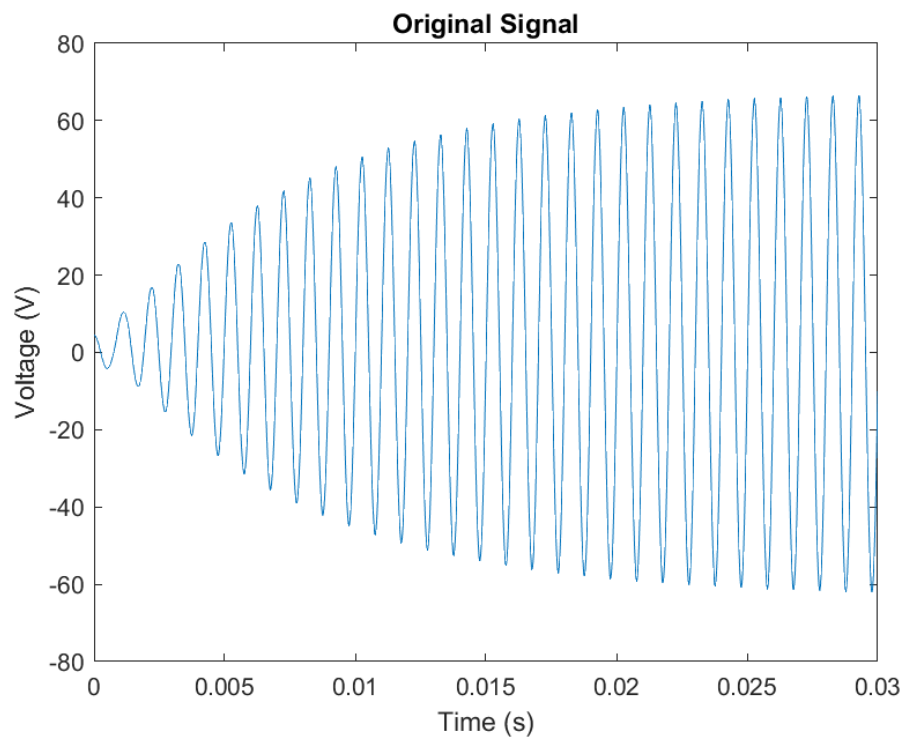


Figure 11: Original wave plotted on MATLAB

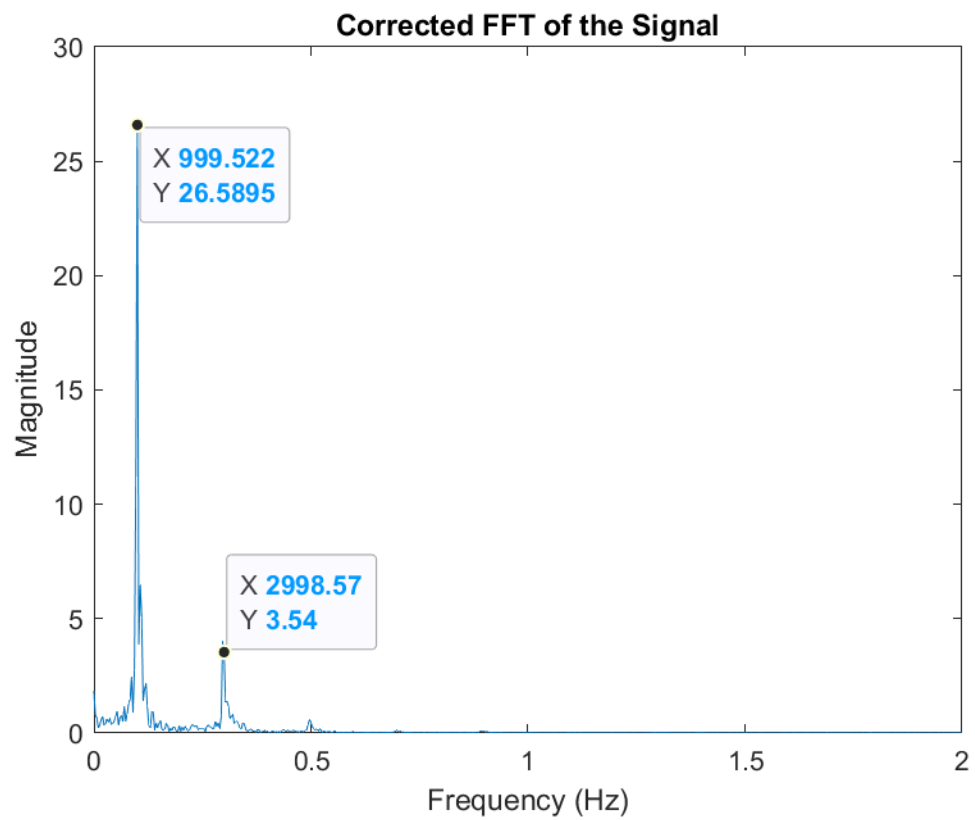


Figure 12: FFT of above wave in MATLAB

This confirms that the main signal component is at 1kHz and smaller peaks at multiples of 1kHz. But why does this occur?

This is due to the setup of this circuit and the main component used, the timer 555. The timer 555 is not a pure sine wave generator. The timer 555 generates a square wave which is composed of multiple frequencies other than the fundamental frequency which is 1kHz. Therefore, unless a lowpass filter is used to reduce the distortion, the timer 555 is not a perfect sine wave generator.